



U32-C (PS2251-17) Specification

Version 1.4

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Revision History

Revision	History	Draft Date	Remark
1.0	First version	March 16, 2021	Drew
1.1	Add flash V6 performance	Nov 19, 2021	Drew
1.2	Update the note of performance	Mar 8, 2022	Drew
1.3	Add N48 performance	Sep 23, 2022	Drew
1.4	Add Bics5 performance	Sep 01, 2023	Yiting

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1. INTRODUCTION



1.1. Product Overview

Phison PS2251-17 U32-C PCBA is a USB Gen 2x1 external storage. It supports USB Gen2, 10Gbps and backward compatible with USB 3.2 Gen1, USB 2.0 and USB 1.1. Product density is up to 4TB. Sequential read performance is up to 1.0GB/s and sequential write performance is up to 800MB/s.

1.2. Block Diagram

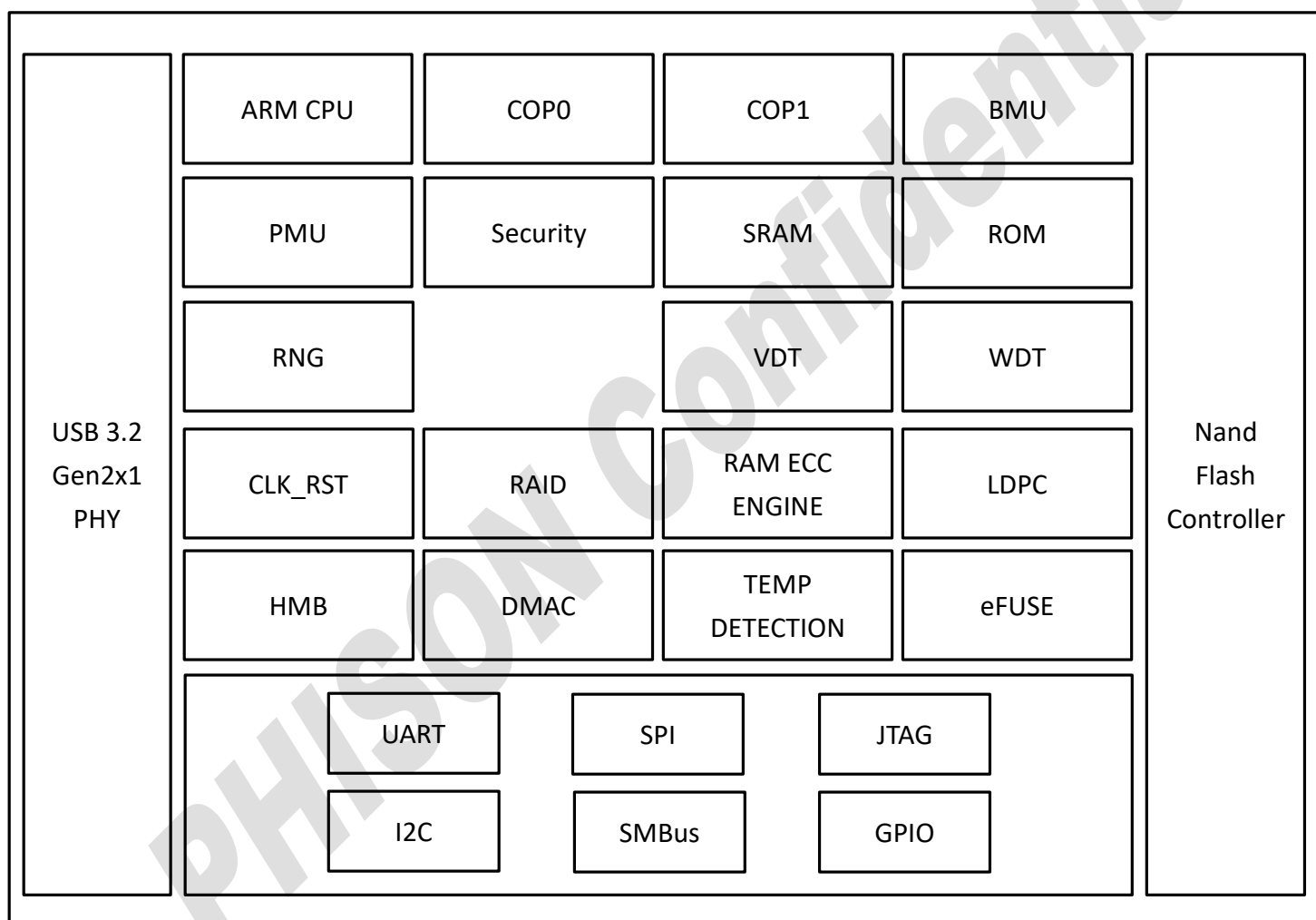


Figure 1-1 System Block Diagram

*PMU: Power Management Unit

1.3. Features Specifications

Features	Specifications
USB	• USB 3.2 Gen 2x1 (10Gbps) • Compliance with USB 3.2 Gen1 (5Gbps) • Compliance with USB 2.0 (480Mbps) • Compliance with USB 1.1 (12Mbps)
NAND Flash	• Support up to 2TB • Support Triple-level cell (TLC) and Quad-level cell (QLC) NAND flash memory • Compliant with Toggle 1.0/2.0/3.0/4.0 NAND flash interface • Compliant with ONFI 2.0/3.0/4.1 • Transfer rate: up to 800 MT/s • Support up to 2 channels/ 16 Chip Enable (CE) within Single Design • Flash IO Operating voltage supply 1.2v/1.8v
Data Reliability	• LDPC ECC protection
OS compatibility	• Support Windows XP and later version • Support MAC OS 10.8 and later version • Support Linux kernel 2.4.10 and later version
Temperature Range	• Operating Range: 0~70 °C • Storage Range: -40~85 °C
Humidity	• 20%~90%
Other Features	• Built-in internal Thermal Sensor

2. ELECTRICAL SPECIFICATIONS



2.1. Operating Voltage

Table 2-1 Operating Voltage

PinName	Min Voltage	Normal Voltage	Max Voltage	Description
AVDD_PAD*	1.8	1.83	1.87	NXTAL Power
AVDDT_PAD	0.915	0.93	0.97	Phy Core Power
V18_REG_PAD	1.7	1.85	1.98	Analog Power Source
VCC18_PAD	1.8	1.83	1.87	Phy Analog Power
VCC3IO_B_PAD	1.75	1.8	1.95	NAND VCCQ Power
	1.19	1.23	1.25	
VCC3IO_PAD	1.75	1.8	1.95	NAND VCCQ Power
	1.19	1.23	1.25	
VCC3IO_ZQ_PAD	1.75	1.8	1.95	NAND VCCQ Power
	1.19	1.23	1.25	
VCCA_H_FC_PAD	2.4	3.3	3.6	NAND VCC Power
VCCA_H_FP_PAD	1.75	1.8	1.95	Analog Power Source
VCCA_H_OSC_PAD	1.75	1.8	1.95	Analog Power Source
VCCA_H_PAD	1.75	1.8	1.95	Analog Power Source
VCCIO0_PAD	1.75	1.8	1.95	GPIO Power
VCCIO1_PAD	1.75	1.8	1.95	GPIO Power
VCCIO2_PAD	1.75	1.8	1.95	GPIO Power
VCK_CORE_PAD	0.915	0.93	0.97	CTL Core Power
VCCQ_P_PAD	1.75	1.8	1.95	Phy Core Power Source
	1.19	1.23	1.25	
VD18_PAD*	1.8	1.83	1.87	USB Analog Power
VD33_PAD*	3	3.3	3.6	USB Analog Power
VDDI33A_PAD	2.35	3.3	3.6	Analog Power Source
VPGM_PAD	1.8	1.83	1.87	Fuse Program Power
VPHY_REG_PAD	0.915	0.93	0.97	Phy Core Power
VREFB_PAD	0.8	0.9	1	Flash channel B Vref

	0.5	0.6	0.7	
VREF_PAD	0.8	0.9	1	Flash channel A Vref
	0.5	0.6	0.7	

NOTE: The values are measurement values, and may vary for difference type of flash and mode of access.

2.2. GPIO DC Characteristic

Pin	Setting			
SONOF	1	0	1	0
CONOF	0	1	1	0
Trigger Voltage	Schmitt Trigger $V_{ih}=0.65*V_{CC}$ $V_{il}=0.35*V_{CC}$	CMOS Trigger $V_{ih}=V_{il}=0.5*V_C$	Two Trigger $V_{ih}=0.52*V_{CC}$ $V_{il}=0.47*V_{CC}$	D=0 (Gating when PAD is floating)

Pin		Setting				
EN		0	1			
IODRV0		X	0	1	0	1
IODRV1		X	0	0	1	1
VCC=1.8V (PMODE=0)	Driving Impedance	Infinity	100	50	33	25
VCC=1.2V (PMODE=1)			$V_o = 0.5*V_{CC}$			

Pin	Setting					
EN	1	0				
PU75	X	0	0	1	0	0
PU5P5	X	0	1	0	0	0
PD75	X	0	0	0	0	1
PD5P5	X	0	0	0	1	0
VCC=1.8V	None	5.5K Pull Up	75K Pull Up	5.5K Pull Up	75K Pull Down	
VCC=1.2V	None	6.5K Pull Up	97K Pull Up	6.5K Pull Up	108K Pull Down	

3. PERFORMANCE



Table 3-1 Performance for Micron N28

N28 (MB/s)		Sequential				Random	
		Read	Write	Sustained Read	Sustained Write	Read	Write
250GB	2CE	1000	320	220	20	150	260
500GB	4CE	1100	700	220	40	200	280
1000GB	8CE	1100	1000	220	40	200	280

Table 3-2 Performance for Hynix V6

V6 (MB/s)		Sequential				Random	
		Read	Write	Sustained Read	Sustained Write	Read	Write
250GB	4CE	1100	700	300	200	200	280
500GB	4CE	1100	700	300	200	200	280
	8CE	1100	1000	300	200	200	280
1000GB	8CE/16CE	1100	1000	300	200	200	280
2000GB	16CE	1100	1000	300	200	200	280

Table 3-3 Performance for Micron N48

N48 (MB/s)		Sequential				Random	
		Read	Write	Sustained Read	Sustained Write	Read	Write
250GB	2CE	1100	700	220	20	150	260
500GB	4CE	1100	1000	220	30	200	280
1000GB	8CE	1100	1000	220	40	200	280
2000GB	16CE	1100	1000	220	40	200	280

Table 3-4 Performance for Kioxia Bics5

Bics5 (MB/s)		Sequential				Random	
		Read	Write	Sustained Read	Sustained Write	Read	Write
250GB	4CE	1000	800	220	100	200	280
500GB	4CE	1000	800	220	100	200	280
1000GB	8CE	1100	1000	220	100	200	280
2000GB	16CE	1100	1000	220	100	200	280

Note:

Performance is measured with the following conditions:

- (a) Sequential : CrystalDiskMark 7.0, 1GB range, QD=Q8T1, Random : CrystalDiskMark 7.0, 1GB range, QD=32T16,
Sustained: H2
- (b) Base on USB 3.2 Gen2x2 (20gbp/s) platform
- (c) Motherboard : X570

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POWER COMSPUTION



Table 4-1 Power Consumption

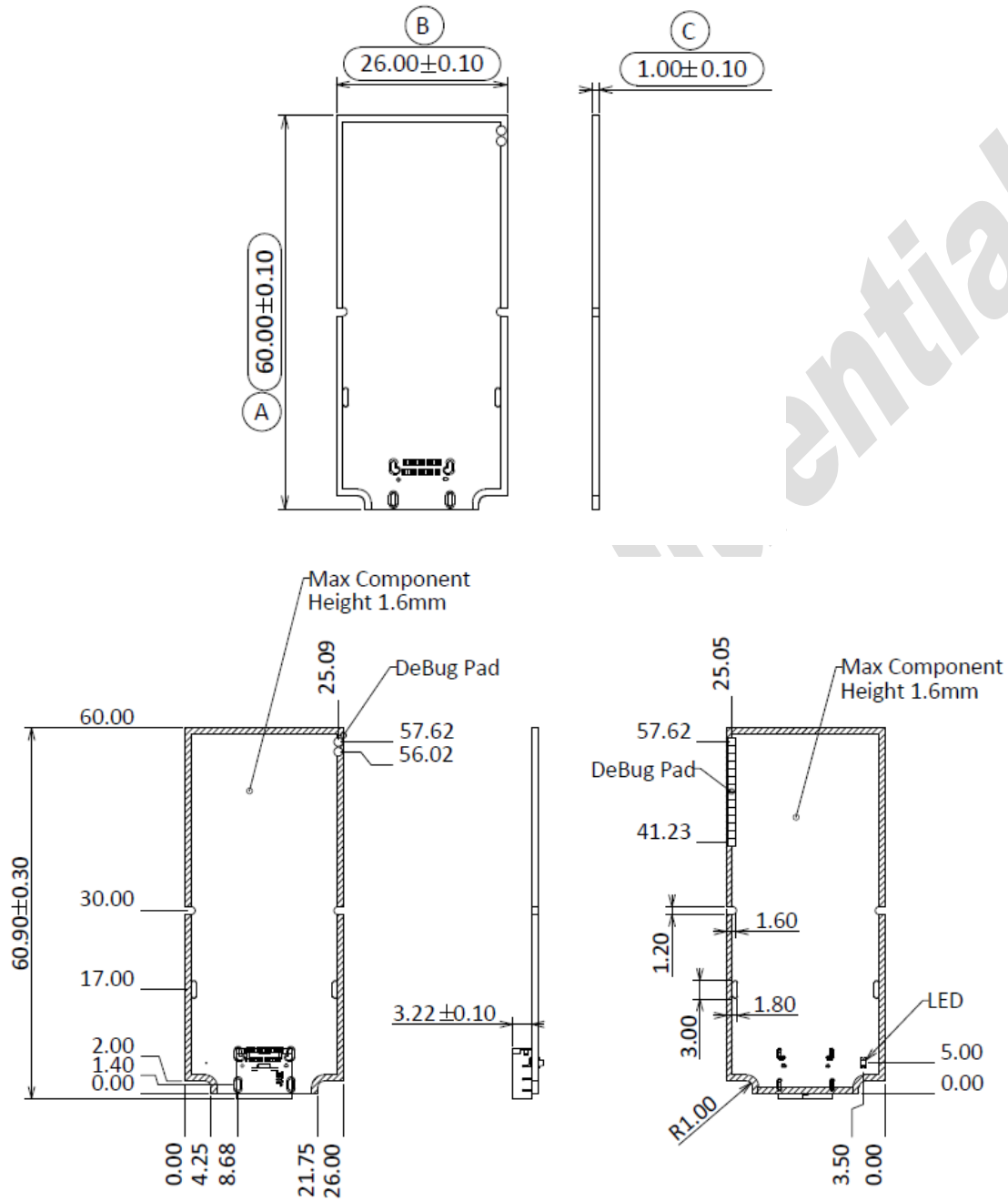
Item	Power (mA)
Read	500
Write	550
Idle	150
Suspend	1.7

Note: Reference only

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4. DIMENSION

- Dimension: 60 ± 0.10 mm (L) x 26 ± 0.10 mm (W) x 1 ± 0.10 mm (H)



Notes :

1. = Max Component Height
2. = No Component
3. = No Component / Signal Vias / Signal Copper / Printing
4. General Tolerance ± 0.1
5. are critical dimensions (A) ~ (C)
6. V-Cut area allowed burr : 0.1mm(max.)
7. SMT Rounter area allowed tolerance: ± 0.05 mm

NOTES:

1. All dimensions are in mm.

5. PRODUCT WARRANTY POLICY



In the event the Product does not conform to the specification within Phison agreed warranty period and such inconformity is solely attributable to Phison's cause, Phison agrees at its discretion replace or repair the nonconforming Product. Notwithstanding the foregoing, the aforementioned warranty shall exclude the inconformity arising from, in relation to or associated with:

- (1) alteration, modification, improper use, misuse or excessive use of the Product;
- (2) failure to comply with Phison's instructions;
- (3) Phison's compliance with customer (including customer's suppliers, subcontractors or downstream customers) indicated instructions, technologies, designs, specifications, materials, components, parts;
- (4) combination of the Product with other materials, components, parts, goods, hardware, firmware or software not developed by Phison; or
- (5) other error or failure not solely attributable to Phison's cause (including without limitation, normal wear or tear, manufacturing or assembly wastage, improper operation, virus, unauthorized maintenance or repair).

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6. REFERENCE



The following table is to list out the standards that have been adopted for designing the product.

Table 7-1 List of References

Title	Acronym/Source
USB 3.2	https://www.usb.org/

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7. TERMINOLOGY



The following table is to list out the acronyms that have been applied throughout the document.

Table 8-1 List of Terminology

Term	Definitions
USB	Universal Serial Bus
SSD	Solid state disk
LDPC	Low Density Parity Check
CE	Chip Enable

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